

LTRS: Improving Word Sense Disambiguation via Learning to Rank Senses

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Abstract

Word Sense Disambiguation (WSD) is a fundamental task critical for accurate semantic understanding. Conventional training strategies usually only consider predefined senses for target words and learn each of them from relatively limited instances, neglecting the influence of similar ones. To address these problems, we propose the method of **Learning to Rank Senses (LTRS)** to enhance the task. This method helps a model learn to represent and disambiguate senses from a broadened range of instances via ranking an expanded list of sense definitions. By employing LTRS, our model achieves a SOTA F1 score of 79.6% in Chinese WSD and exhibits robustness in low-resource settings. Moreover, it shows excellent training efficiency, achieving faster convergence than previous methods. This provides a new technical approach to WSD and may also apply to the task for other languages¹.

1 Introduction

Word Sense Disambiguation (WSD) aims to identify the sense of words in context (Navigli, 2009), which is critical for accurate semantic understanding and beneficial to multiple downstream applications, such as Information Retrieval (Biloshmi et al., 2021), Text Summarization (Kouris et al., 2021), Machine Translation (Emelin et al., 2020). In recent years, integrating lexical knowledge, such as sense definitions, within neural architectures has successfully enhanced the performance of supervised WSD methods (Huang et al., 2019; Blevins and Zettlemoyer, 2020; Barba et al., 2021a,b).

Although these methods have achieved decent results, a significant performance drop remains between the most frequent senses (MFS) and the less frequent senses (LFS). This can be attributed to the imbalance in training data, where LFS are seldom

represented as positive senses, hindering effective learning for them. Some methods have attempted to tackle this problem by specifically annotating more instances for LFS (Blevins et al., 2021) or balancing the learning between MFS and LFS by loss reweighting (Su et al., 2022). However, it is labor-intensive and time-consuming to acquire instances for rare senses, while loss reweighting may lead to overfitting due to insufficient data for LFS.

We observe that, from a linguistic perspective, words holding similar senses tend to appear in similar contexts, indicating that instances for a sense may benefit the learning of similar senses. For example, as shown in Table 1, the sense "宽阔₁ ((area) wide)" and "坦荡₁ (wide and flat)", "宽阔₂ (not narrow-minded)" and "坦荡₂ (pure and open-hearted)" are close with similar contexts for exploration. This phenomenon is also evident in other languages, such as "wide" and "broad" in English, as shown in Table 2. This again suggests that leveraging sense similarity may enhance WSD.

However, conventional training strategies may neglect the influence of similar senses (Erk et al., 2013), as they usually only consider predefined senses for target words and treat all of them equally. To address these problems, Learning to Rank (LTR) methods, widely applied in fields such as Recommendation Systems (Karatzoglou et al., 2013) and Information Retrieval (Liu et al., 2009), may be helpful. By employing LTR, models can effectively distinguish among highly similar, moderately similar, and dissimilar objects to a given query. Compared to query-object pairs used in the above-mentioned fields, this may also apply to word-definition pairs needed in WSD scenarios.

Based on these considerations, we are motivated to enhance WSD by adjusting the learning process and propose the method of **Learning to Rank Senses (LTRS)**. At training time, the model is encouraged to rank sense definitions according to their semantic similarity with the target word. Ad-

¹The Code for this paper is available at <https://github.com/COOLPKU/LTRS>.

Word	Sense ID	Sense Definition	Context
宽阔	宽阔 ₁	面积广 (area) wide	~的河面在阳光下闪闪发光。 The ~ river surface sparkled in the sunlight.
	宽阔 ₂	心地开阔不狭隘 not narrow-minded	他有着~的胸襟，能容纳不同的意见。 He has a ~ mind that can accommodate different opinions.
坦荡	坦荡 ₁	宽广平坦 wide and flat	~的大路在阳光的照耀下延伸到远方。 The ~ road stretches into the distance under the shining sun.
	坦荡 ₂	心地纯洁宽畅 pure and open-hearted	他的心胸~，能够包容别人的过错。 His mind is ~, capable of forgiving others' mistakes.

Table 1: Senses for "宽阔" and "坦荡", from the sense inventory WrdInv of MiCLS (Wang et al., 2024).

Word	Sense ID	Sense Definition	Context
wide	wide ₁	measuring a lot from one side to the other	The river was so ~ that it took an hour to row across it.
	wide ₂	including a large number or variety of things	The store offers a ~ range of goods.
broad	broad ₁	wide	The ~ street was lined with trees.
	broad ₂	including a great variety of things	The company has a ~ range of products.

Table 2: Senses for "wide" and "broad", from Oxford English Dictionary (Dictionary, 1989).

ditionally, the candidate definition list is expanded by including definitions from other words. In this way, the model can learn to represent and disambiguate senses from a broadened range of instances, which is especially helpful for LFS.

By employing LTRS, our model outperforms previous top-performing models in Chinese WSD and exhibits robustness in low-resource settings. Furthermore, it also achieves better training efficiency than the previous. Considering the generality of these linguistic issues, this method may also apply to the task for other languages.

2 Related Works

WSD Methods: Recent supervised neural WSD methods have achieved decent performance by leveraging lexical knowledge bases (Bevilacqua et al., 2021), with incorporating definitional (Huang et al., 2019; Blevins and Zettlemoyer, 2020; Barba et al., 2021a,b), relational (Vial et al., 2019; Bevilacqua and Navigli, 2020; Wang and Wang, 2020; Song et al., 2021; Zhang et al., 2022), and morphological (Zheng et al., 2021; Wang et al., 2024) knowledge. Some methods have further improved the performance on LFS by annotating more instances for them (Blevins et al., 2021) or adopting Z-reweighting (Su et al., 2022). However, these methods only consider predefined senses for target words and learn each from relatively limited instances, potentially preventing models from fully leveraging sense similarity.

LTR Methods: Existing methods fall into three categories: Pointwise methods (Crammer and Singer,

2001; Li et al., 2007) independently optimize the similarity score of each query-object pair ignoring relationships between objects; Pairwise methods (Burges et al., 2005; Cao et al., 2006) improve it by modeling preferences between two objects but overlook their global positions; Listwise methods (Cao et al., 2007; Xia et al., 2008) focus on the overall order of all objects rather than individual objects or pairs. Listwise methods are more appropriate for WSD than the others, since the most suitable sense needs to be identified while ensuring similar ones with relatively high global positions.

3 Methodology

3.1 Task Formulation

We frame WSD as a multi-class classification task. Given a polysemous word w in context c_w , a WSD system needs to identify the most suitable sense definition from $D_w = \{d_i\}_{i=1}^l$, the sense definition set for w . To find this targeting definition, our method requires a function f for mapping a (w, d) pair to a similarity score s . At prediction time, the most suitable definition for w is determined as:

$$\hat{d} = \arg \max_d f(w, d), \text{ where } d \in D_w. \quad (1)$$

3.2 WSD Enhanced by LTRS

The general idea of LTRS is to help a model learn to rank definitions based on their semantic similarity with the target word. The overall architecture of our method is shown in Figure 1. Specifically, given a mini-batch of target words $W = \{w_i\}_{i=1}^m$ and corresponding contexts $C_W = \{c_{w_i}\}_{i=1}^m$, we devise a

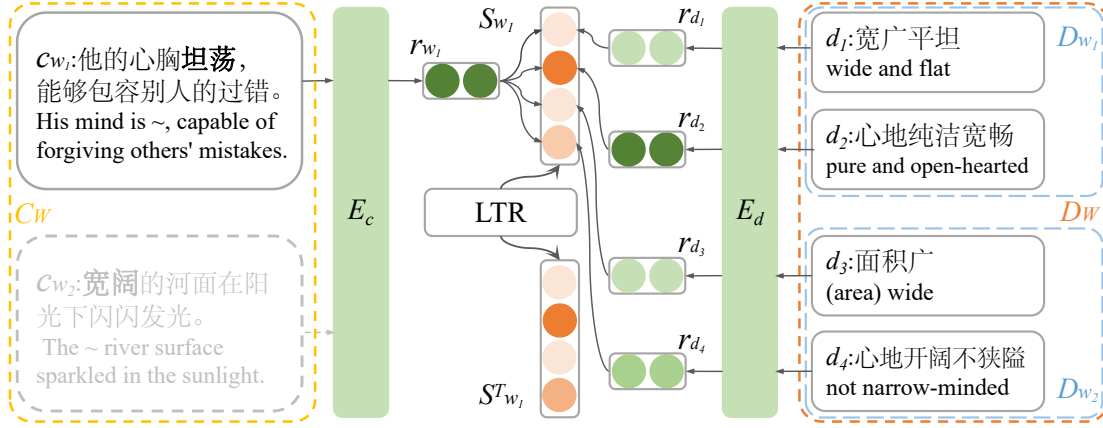


Figure 1: Illustration of the proposed LTRS: the model is required to rank an expanded list of sense definitions according to their semantic similarity with the target word (shown in **bold**). In this way, it learns to represent and disambiguate senses of target words from instances for similar senses, such as "宽阔" and "他的心胸~, 能够包容别人的过错 (His mind is ~, capable of forgiving others' mistakes)".

unified definition set $D_W = \bigcup_{i=1}^m D_{w_i}$. A context encoder E_c and a definition encoder E_d are used to get the representation $\mathbf{r}_w$ for each $w \in W$ and $\mathbf{r}_d$ for each $d \in D_W$, respectively. Both encoders are initialized with the pre-trained model BERT (Devlin et al., 2019). Their inputs are padded with the BERT-specific classification token [CLS] and separator token [SEP]. In addition, the target word in context inputs is replaced by a [MASK] to enhance generalization. We obtain $\mathbf{r}_w$ from the output embedding of [MASK] and $\mathbf{r}_d$ from that of [CLS].

For each $w \in W$, the predicted score list of candidate definitions is defined as $S_w = [\phi(\mathbf{r}_w, \mathbf{r}_{d_i})]_{i=1}^{|D_W|}$, where $\phi(\mathbf{r}_w, \mathbf{r}_{d_i}) = \frac{\mathbf{r}_w \cdot \mathbf{r}_{d_i}}{\|\mathbf{r}_w\| \|\mathbf{r}_{d_i}\|}$ and $d_i \in D_W$.

To evaluate S_w , we compare it with the ground truth score list S_w^T . Based on the semantic equivalence between w and the correct definition d^* , the ground truth score for (w, d) can be measured by the similarity score between d^* and d . To compute the similarity score, we apply a sentence embedding model BGE (Xiao et al., 2024), which has achieved SOTA performance on many Semantic Textual Similarity (STS) tasks. Formally, $S_w^T = [\phi(E(d^*), E(d_i))]_{i=1}^{|D_W|}$, where E is a BGE encoder.

To help the model learn ranking knowledge from the ground truth scores, two listwise LTR methods ListNet (Cao et al., 2007) and ListMLE (Xia et al., 2008) are utilized:

ListNet aims to minimize the cross entropy between the top one probability² distribution and the

ground truth. Given the score list of all definitions $S = [s_i]_{i=1}^n$, the top one probability of definition i represents the probability of its being ranked at top-1, calculated as: $P_S(i) = \frac{e^{s_i/\tau}}{\sum_{j=1}^n e^{s_j/\tau}}$, where τ is a temperature hyperparameter for smoothing the distribution. The objective of ListNet is defined as:

$$\mathcal{L}_{\text{ListNet}} = - \sum_{i=1}^{|S_w|} P_{S_w^T}(i) \log P_{S_w}(i). \quad (2)$$

We use different temperature hyperparameters for S_w and S_w^T , denoted by τ_1 and τ_2 .

ListMLE aims to maximize the log-likelihood of the ground truth permutation for the definition indexes $\pi^T = [\pi^T(i)]_{i=1}^{|S_w^T|}$, which represents definition $\pi^T(i)$ is ranked i -th. The objective of ListMLE is defined as³:

$$\mathcal{L}_{\text{ListMLE}} = - \log \prod_{i=1}^k \frac{e^{s_{\pi^T(i)}/\tau_3}}{\sum_{j=i}^{|S_w^T|} e^{s_{\pi^T(j)}/\tau_3}}, \quad (3)$$

where $s_{\pi^T(i)} \in S_w$, τ_3 is a temperature hyperparameter, $k (< |S_w^T|)$ is a hyperparameter for efficiency consideration⁴.

4 Experiment and Analysis

4.1 Experimental Settings

Datasets: We fuse FiCLS (Zheng et al., 2021) and MiCLS (Wang et al., 2024) to increase the data volume consideration.

³In practice, in order for the model to place greater emphasis on higher-ranked senses, the losses for higher rankings are assigned higher weights based on S_w^T .

⁴The original permutation probability is calculated with $k = |S_w^T|$. Since we mainly focus on the top few closest senses, k is introduced to reduce computational complexity.

²ListNet can be based on either permutation probability or top one probability. We adopt the top one probability for

	Valid	Test				
		Noun	Verb	Adj.	Adv.	ALL
MFS	42.3	46.2	39.9	39.9	32.5	42.1
BERT	73.2	74.7	72.5	74.4	71.1	73.2
GlossBERT	75.8	75.6	76.2	75.6	73.3	75.5
BEM	78.3	78.4	78.7	78.1	73.2	78.1
FormBERT	77.6	77.2	77.4	78.8	75.5	77.4
ESCHER	78.3	76.7	78.9	79.6	75.6	77.9
LTRS _{ListNet}	80.2	78.5	80.8	80.6	77.3	79.6
LTRS _{ListMLE}	79.7	78.6	80.2	80.0	78.5	79.3

Table 3: Comparison of F1 scores (%) for Chinese WSD. The best results are shown in **bold**.

ume and sense coverage, providing more effective training and validation for our method. The new dataset contains 96829 instances, covering 88.1% of polysemous words and 77.9% of their senses in CCD⁵. We divide the new dataset into training, validation, and test sets by 7:1:2. More details about this sourced dataset are shown in Appendix A.

Baselines: Besides MFS and BERT (Devlin et al., 2019) as default baselines, we compare LTRS with four recent top-performing systems⁶, including GlossBERT (Huang et al., 2019), BEM (Blevins and Zettlemoyer, 2020), FormBERT (Zheng et al., 2021) and ESCHER (Barba et al., 2021a), with the same settings as our model for a fair comparison.

Experimental Configuration: We adopt chinese-bert-base-wwm-ext (Cui et al., 2020) as the base model and bge-large-zh-v1.5 (Xiao et al., 2024) for computing the ground truth scores. The settings of the two models and other detailed configurations are shown in Appendix B.

4.2 Evaluation Results

Overall Results: Table 3 shows the overall results for Chinese WSD across the main parts-of-speech (PoS). From it, we have the following observations:

(1) By LTRS, our model achieves the best F1 score of the test set and surpasses all competitors across all PoS. Compared to BEM, LTRS_{ListNet} and LTRS_{ListMLE} outperforms it by 1.5 and 1.2 F1 points respectively with the same bi-encoder architecture. This can be attributed to the enhanced learning of sense representation and disambiguation from a broadened range of instances via ranking an expanded list of sense definitions.

⁵CCD is the abbreviation of the Contemporary Chinese Dictionary, the most authoritative Chinese dictionary.

⁶Other top-performing WSD methods, such as EWISER (Bevilacqua and Navigli, 2020) and ConSeC (Barba et al., 2021b), require special word features that are unavailable in existing Chinese WSD datasets.

	MFS	LFS	Zero-shot
BEM	86.3	71.6	62.3
ESCHER	87.1	70.8	57.6
LTRS _{ListNet}	85.6	75.3	70.0
LTRS _{ListMLE}	85.9	74.6	69.3

Table 4: Comparison of LTRS against its competitors on MFS, LFS, and Zero-shot subsets of the test set.

t	Instances
1	19277
3	43844
5	55883
unlimited	67780

Table 5: Number of training instances at different values of t .

(2) LTRS_{ListNet} and LTRS_{ListMLE} achieve relatively consistent results across all sets and PoS, validating the effectiveness of both LTR methods.

Results in Low-resource Settings: To better understand the overall results, we also consider three subsets of the test set: (i) instances for MFS, (ii) instances for LFS, and (iii) zero-shot instances for unseen senses during training. As shown in Table 4, LTRS introduces significant improvements over its competitors on LFS and Zero-shot, validating the robustness of our method in low-resource settings. This is also due to our method’s ability of learning senses from other instances in the mini-batch. Despite slightly lower performance on MFS, its exceptional capability in low-resource scenarios contributes to the improvements on the whole.

Separate Results on FiCLS and MiCLS: To address the problem of reproducibility and comparison with other papers, we conduct separate training and evaluation on MiCLS and FiCLS. Detailed results are shown in Appendix C, which indicate that LTRS still outperforms its competitors and demonstrate the effectiveness of our method.

Case Study: A case study (detailed in Appendix D) is conducted to further explore the reasons for the promising performance of LTRS, which shows that it can leverage definitional knowledge and instances more fully and effectively.

4.3 Few-Shot Evaluation

We compare LTRS and BEM in a few-shot scenario with $t \in \{1, 3, 5, \text{unlimited}\}$ training instances per sense. The number of instances during training for each t is shown in Tabel 5.

As shown in Figure 2, all models achieve better

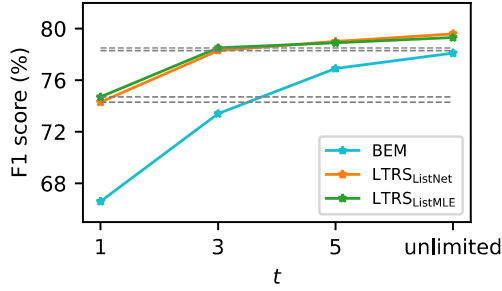


Figure 2: F1 scores (%) of BEM and LTRS on the test set, when varying t .

	Test
BEM w/ BERT-base	78.1
BEM w/ BGE-base	78.7
BEM w/ BGE-large	79.8
LTRS _{ListNet} w/ BERT-base	79.6
LTRS _{ListMLE} w/ BERT-base	79.3

Table 6: F1 scores (%) of LTRS and BEM based on different pre-trained models.

F1 scores as t increases. However, LTRS makes more efficient use of training data and achieves similar results to the strongest BEM with only 3 instances per sense.

4.4 Analysis on the Contribution of BGE

To analyze the contribution of BGE apart from LTRS, we conduct additional experiments to evaluate the performance of BGE-based BEM. Two BGE models with different sizes are employed: bge-base-zh-v1.5 and bge-large-zh-v1.5 (Xiao et al., 2024). Detailed settings of them are provided in Table 10. As shown in Table 6, LTRS significantly outperforms BEM based on BGE-base, indicating that the performance gains are primarily due to the LTR strategy. Notably, LTRS achieves performance comparable to BEM based on BGE-large while using only approximately one-third of its fine-tuned parameter quantity.

4.5 Analysis on Batch Size

We conduct additional experiments to investigate the impact of the definition batch size. Results on the test set for various batch sizes are presented in Table 7, showing that the larger the batch size, the better LTRS performs. A possible reason for this is that with a larger batch size, the model can learn more similarity knowledge between senses. However, its setting is constrained by memory.

	Batch Size		
	64	128	256
LTRS _{ListNet}	79.2	79.5	79.6
LTRS _{ListMLE}	78.7	79.3	79.3

Table 7: Comparison of different batch sizes for LTRS.

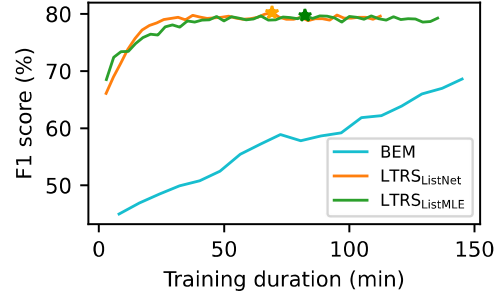


Figure 3: Training curves of BEM and LTRS. The best performance of each model is denoted by a star.

4.6 Analysis on Training Efficiency

We further compare LTRS and BEM on training efficiency, with the same experimental settings introduced in Appendix B. Figure 3 shows that our model achieves the best validation performance within 100 minutes, noticeably faster than BEM. This can be attributed to the varying number of senses for each word, which limits BEM’s parallel processing. In our method, a unified sense definition set is devised to effectively tackle this issue. Compared to BEM’s 24.2 minutes per epoch, LTRS_{ListNet} and LTRS_{ListMLE} need only 9.6 and 9.8 minutes, respectively. In addition, LTRS provides more learning opportunities per epoch for the senses, which helps to accelerate convergence.

5 Conclusion

In this paper, we propose the LTRS method to enhance WSD. By ranking an expanded list of sense definitions, the model can learn to represent and disambiguate senses from a broadened range of instances. Our model achieves a SOTA F1 score of 79.6% in Chinese and exhibits robustness in low-resource settings. Moreover, it shows excellent training efficiency, achieving faster convergence than previous methods.

This method provides a novel technical approach to WSD. In the near future, we will go further to evaluate it in more languages, particularly focusing on low-resource settings such as LFS, few-shot, and zero-shot, considering that manually annotated data may be relatively scarce in some low-resource languages.

6 Limitations

Despite achieving promising results, there remain some limitations of our method as follows:

(1) The superior performance of LTRS is related to the lexical sample WSD datasets we employed, which include a relatively high proportion of LFS and zero-shot senses. Our method evidently excels in disambiguating these types of senses, resulting in significant performance gains on the whole. However, in other benchmarks where the proportions of lower frequent senses are comparatively lower, the advantage of LTRS may be less pronounced.

(2) Our method relies on an extra top-performing sentence embedding model, BGE (Xiao et al., 2024) for example, to compute the similarity scores between sense definitions. In low-resource languages, this kind of sentence embedding model may be less accurate for measuring the similarity between definitions, thereby weakening the effectiveness of LTRS.

(3) Similar to previous methods, our method also fails to achieve significant performance gain on words with fine-grained sense categorization. For this scenario, we conduct a detailed case study in Appendix D. To achieve accurate sense disambiguation at this level of granularity, supplementary lexical semantic and syntactic knowledge may be required.

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A Data Statistics

FiCLS (Zheng et al., 2021) and MiCLS (Wang et al., 2024) are currently the two largest available Chinese lexical sample WSD datasets. Both of them use the CCD-originated sense inventory for annotation. However, FiCLS covers limited disyllabic words while MiCLS only covers disyllabic words. So we combine all data targeting polysemous words⁷ into a new dataset. To unify the sense definitions, all of them are retrieved from WrInv and MorInv provided by MiCLS.

The data statistics for each source are shown in Table 8. The sourced dataset covers 88.1% of polysemous words and 77.9% of their senses in CCD. We divide it into training, validation, and test sets by 7:1:2. The statistics of these sets are shown in Table 9.

Source	Words	Senses	Instances
FiCLS	1888	5997	36698
MiCLS	8126	14948	60131
ALL	10014	20945	96829

Table 8: The data statistics for different sources.

Split	Words	Senses	Instances	Context Length	Definition Length
Train	9812	19277	67780	27.5	11.6
Valid	4883	6323	9683	27.3	11.4
Test	6906	10217	19366	27.6	11.6

Table 9: Statistics of the training, validation, and test sets. The length is calculated as the average number of Chinese characters.

B Experimental Configuration

The settings of the pre-trained models involved in the experiments are shown in Table 10. All of them adopt an architecture based on the Transformer (Vaswani et al., 2017) encoder.

⁷MiCLS covers both monosemous and polysemous words. Although LTRS can leverage monosemous lexical examples during training, we only use polysemous word data for a fair comparison with other methods.

	Model Size	Encoder Layers	Attention Heads	Hidden Size
chinese-bert-base-wwm-ext	110M	12	12	768
bge-base-zh-v1.5	102M	12	12	768
bge-large-zh-v1.5	326M	24	16	1024

Table 10: The settings of BERT and BGE we employ.

We carry out grid search of temperatures $\tau_1, \tau_2, \tau_3 \in \{0.01, 0.05, 0.1, 0.2\}$, $k \in \{3, 5, 10\}$, definition batch size $\in \{64, 128, 256\}$, and select the best combination based on the validation performance. Finally, we set τ_1, τ_2, τ_3 to 0.05, and k to 5. The model is finetuned by AdamW (Loshchilov and Hutter, 2017) optimizer for up to 20 epochs with a learning rate of $5e-5$. Before the beginning of each epoch, we randomly shuffle the training data. We evaluate the model every 250 training steps on the valid set and keep the best checkpoint for evaluation on the test set.

For the BERT baseline, we finetune a linear classifier on the hidden states of the target word output by a frozen BERT. For BEM, the same settings as our model are adopted. For the other baselines, we uniformly adopt the same pre-trained model as LTRS, and directly follow the experimental configurations described in their original papers for the other settings.

All experiments are conducted with the deep learning framework PyTorch on a single NVIDIA RTX 3090 GPU (43GB memory).

C Separate Results on FiCLS and MiCLS

The separate results on FiCLS and MiCLS are shown in Table 11. The results appear to be higher than the fusion results because MiCLS intentionally includes some monosemous word data, which may actually be polysemous in the real corpus. In the fusion, we filter out these data to ensure a fair comparison with other models.

D Case Study

LTRS achieves remarkable performance compared to previous methods, particularly on LFS. To better understand the underlying reasons, we conduct case studies as below:

Take the word "花红" in the context "春节这天，老板要发放年终～ (On the Spring Festival,

	FiCLS					MiCLS				
	Noun	Verb	Adj.	Adv.	ALL	Noun	Verb	Adj.	Adv.	ALL
MFS	35.2	34.5	33.3	36.7	35.0	80.6	76.5	71.7	66.0	77.6
BERT	74.7	71.1	72.1	64.3	71.8	-	-	-	-	-
GlossBERT	82.9	82.0	82.6	81.9	84.5	-	-	-	-	-
BEM	73.2	72.6	74.6	66.2	72.2	88.4	87.9	85.1	76.3	87.4
FormBERT	88.7	87.7	88.5	83.1	87.6	93.0	92.1	88.2	83.5	91.9
MorBERT	-	-	-	-	-	93.2	92.5	88.9	84.0	92.2
LTRS _{ListNet}	88.9	89.4	89.3	85.1	88.2	93.7	93.8	90.2	84.9	93.1
LTRS _{ListMLE}	88.8	89.0	89.5	83.9	88.0	93.8	93.6	90.6	85.8	93.2

Table 11: Separate results on the test set of FiCLS and MiCLS. The results of baselines are sourced from the original papers for the two datasets.

Word	Sense ID	Sense Definition	Context
观览	观览 ₁	参观观看 visit and view	我们前往美术馆～展览，欣赏各种风格迥异的艺术作品。 We went to the gallery to ~ the exhibition, admiring a variety of diverse art pieces.
	观览 ₂	参观游览 visit and tour	游客纷至沓来，～景点，流连忘返。 Tourists flock to ~ attractions, reluctant to leave.
重利	重利 ₁	很高的利息 very high interest	贷款公司因～盘剥而受到广泛批评。 The loan company has been widely criticized for charging ~.
	重利 ₂	很高的利润 very high profit	该公司被指控利用不公平竞争手段牟取～。The company has been accused of using unfair competition tactics to reap ~.
原装	原装 ₁	原来包装好的 originally packaged	在新年宴会上，老板特别准备了几瓶～名酒。 At the New Year's banquet, the boss specially prepared several bottles of ~ brand liquor.
	原装 ₂	原来装配好的 originally assembled	我们购买了～彩电，以享受更清晰的画质和更稳定的功能。 We purchased an ~ television to enjoy clearer picture quality and more stable features.

Table 12: Fine-grained sense categorization for "观览", "重利", and "原装", with extra lexical knowledge and contexts needed for accurate sense disambiguation.

the boss will distribute the year-end ~)" as an example. Our model properly identifies the sense "花红₃ (bonus)" for it, while there are no instances for "花红₃ (bonus)" in the training set. The reason for this is that the model can learn to represent "花红₃ (bonus)" from instances for senses similar to it, such as "红利₂ (rewards given to employees by the company)" and "奖金₁ (money as rewards)". Moreover, when training on these instances, it also learns to differentiate "花红₃ (bonus)" from the other senses, including "花红₁ (a holiday gift)" and "花红₂ (a kind of plant)". This explicit learning helps our model achieve more performance gain. A similar case can also be seen on "解₁ (send away under escort)" and its similar sense "监押₂ (escort under supervision)".

However, similar to previous methods, our method also fails to achieve significant performance gain on words with fine-grained sense categorization. For example, the model misclassifies an instance annotated with the sense "观览₁ (visit

and view)" and assigns "观览₂ (visit and tour)" to it. This is because the two senses are too similar and difficult to distinguish, as shown in Table 12. In this scenario, BGE tends to output very close ground truth scores for both senses, thereby hindering models' learning. Similar cases can also be seen on "重利" and "原装" shown in Table 12. To address this problem, extra lexical knowledge and contexts may be needed for accurate sense disambiguation.